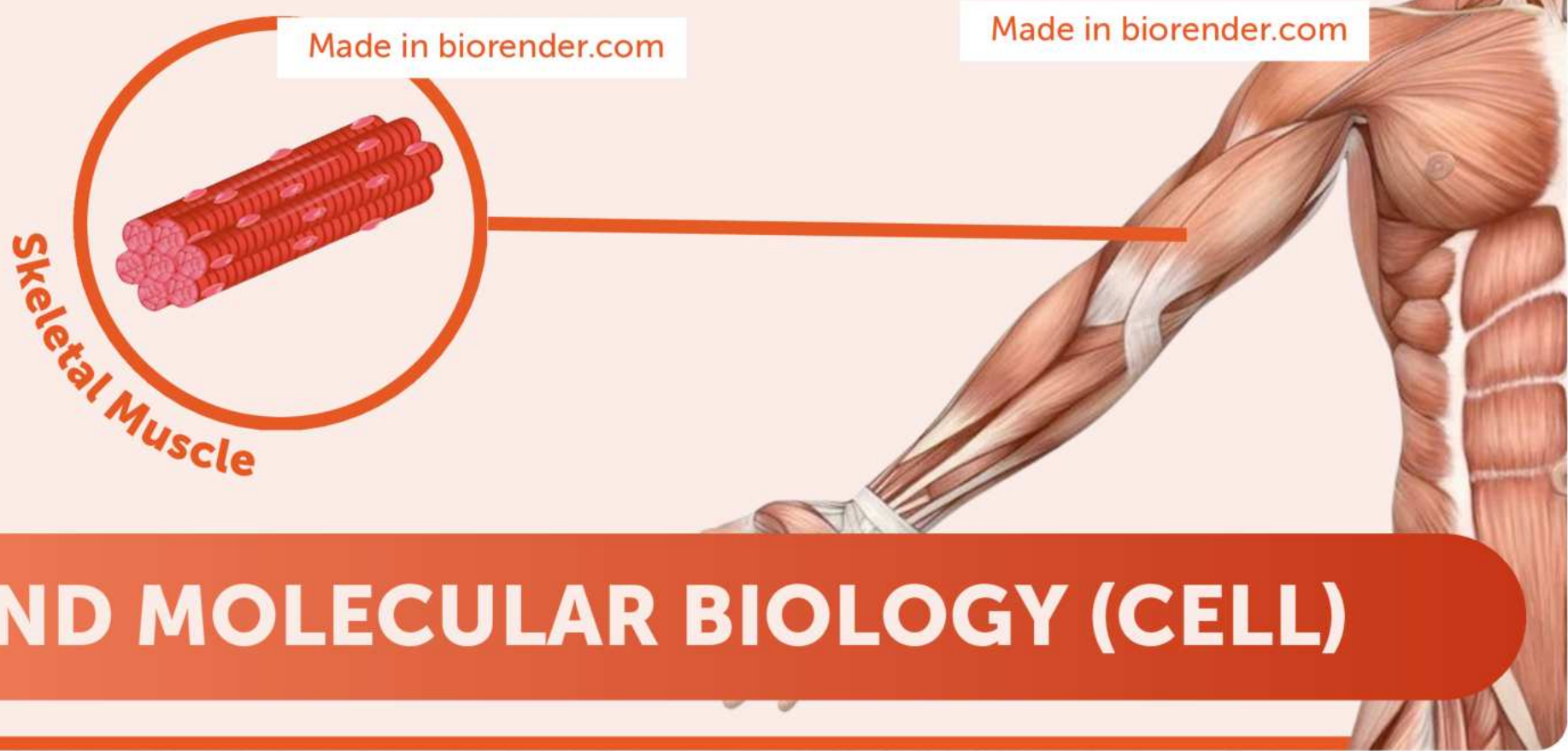


Analyzing the effect of Eucalyptol on a skeletal muscle atrophy model of C2C12 myotubes by measuring the expression of atrophy indicators FBX032 and TRIM63



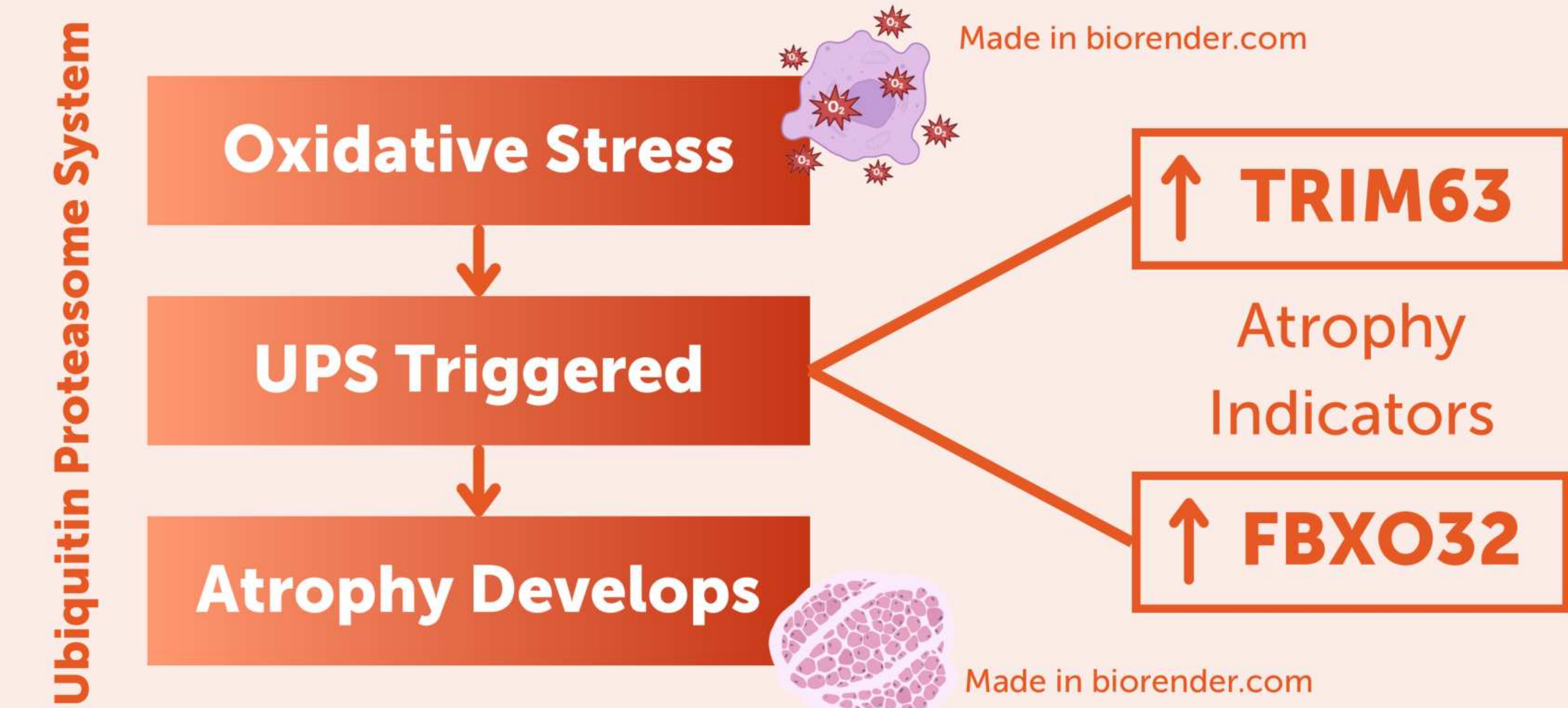
CELLULAR AND MOLECULAR BIOLOGY (CELL)

BACKGROUND

Skeletal Muscle Atrophy (SMA): Skeletal muscle disease characterized by faster protein breakdown than synthesis.



SMA primarily develops by high levels of **oxidative stress** triggering the **Ubiquitin Proteasome System (UPS)**



PROPOSED TREATMENT - EUCALYPTOL

- Strong **antioxidant** properties
- Previous research found treatments with **antioxidant properties** showed **alleviation of SMA** (Hoch et al.)
- Underexplored in SMA
- Current treatments possess side effects and economically unviable
- **Exercise**, a **counterintuitive** approach, is a widely accepted treatment

RESEARCH QUESTION

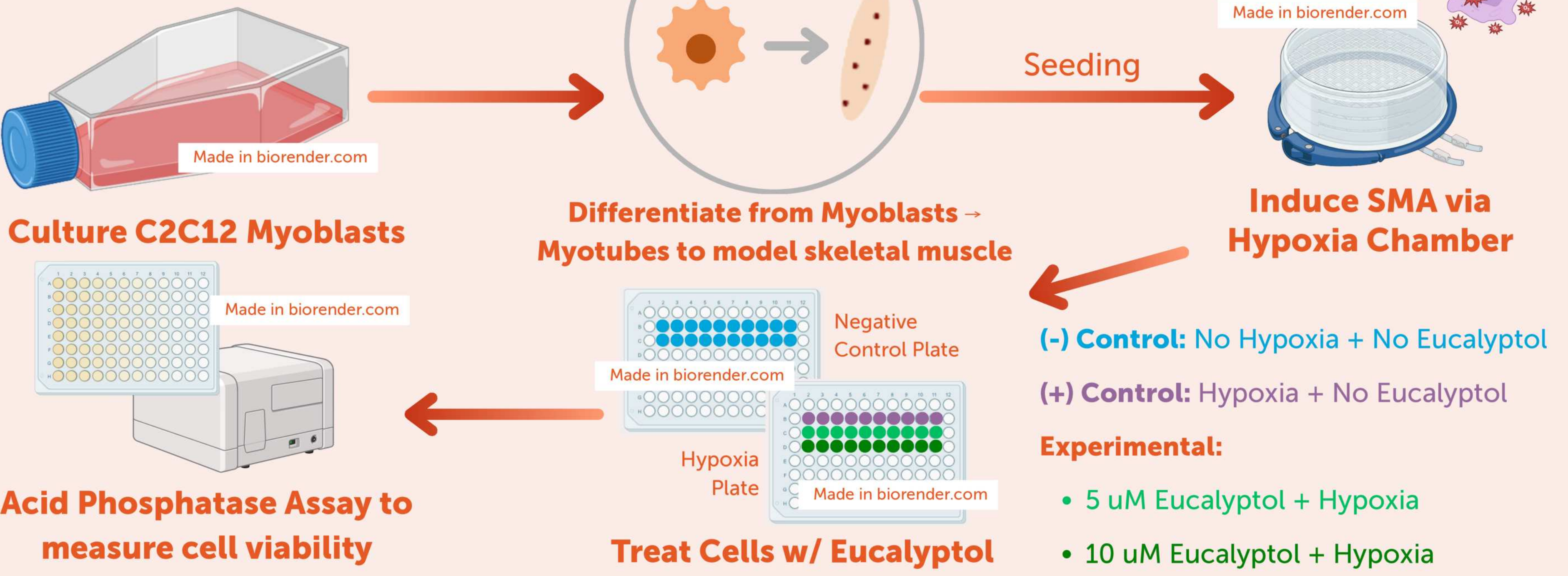
Can Eucalyptol alleviate Skeletal Muscle Atrophy in C2C12 Myotubes by downregulating the gene expression of FBX032 and TRIM63?

HYPOTHESIS

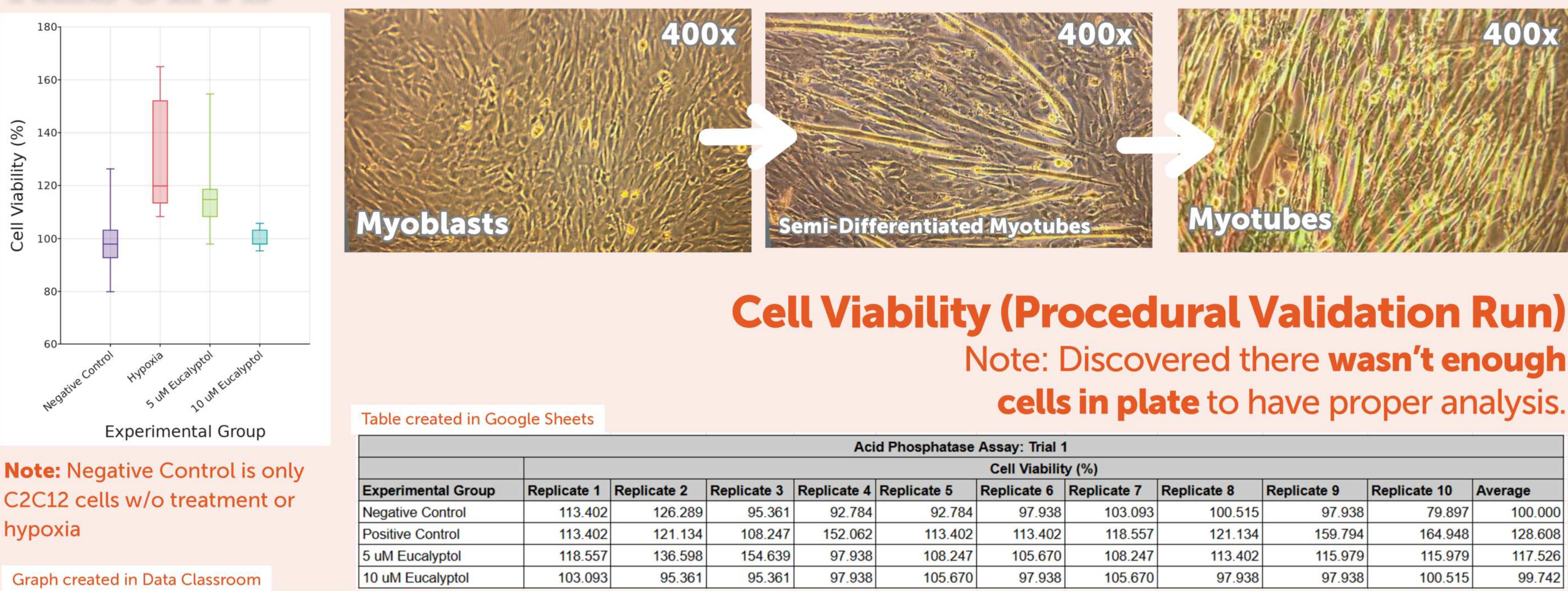
If C2C12 myotubes are treated with Eucalyptol, **then** I will see a **downregulation** of atrophy indicating genes **FBX032** and **TRIM63**.



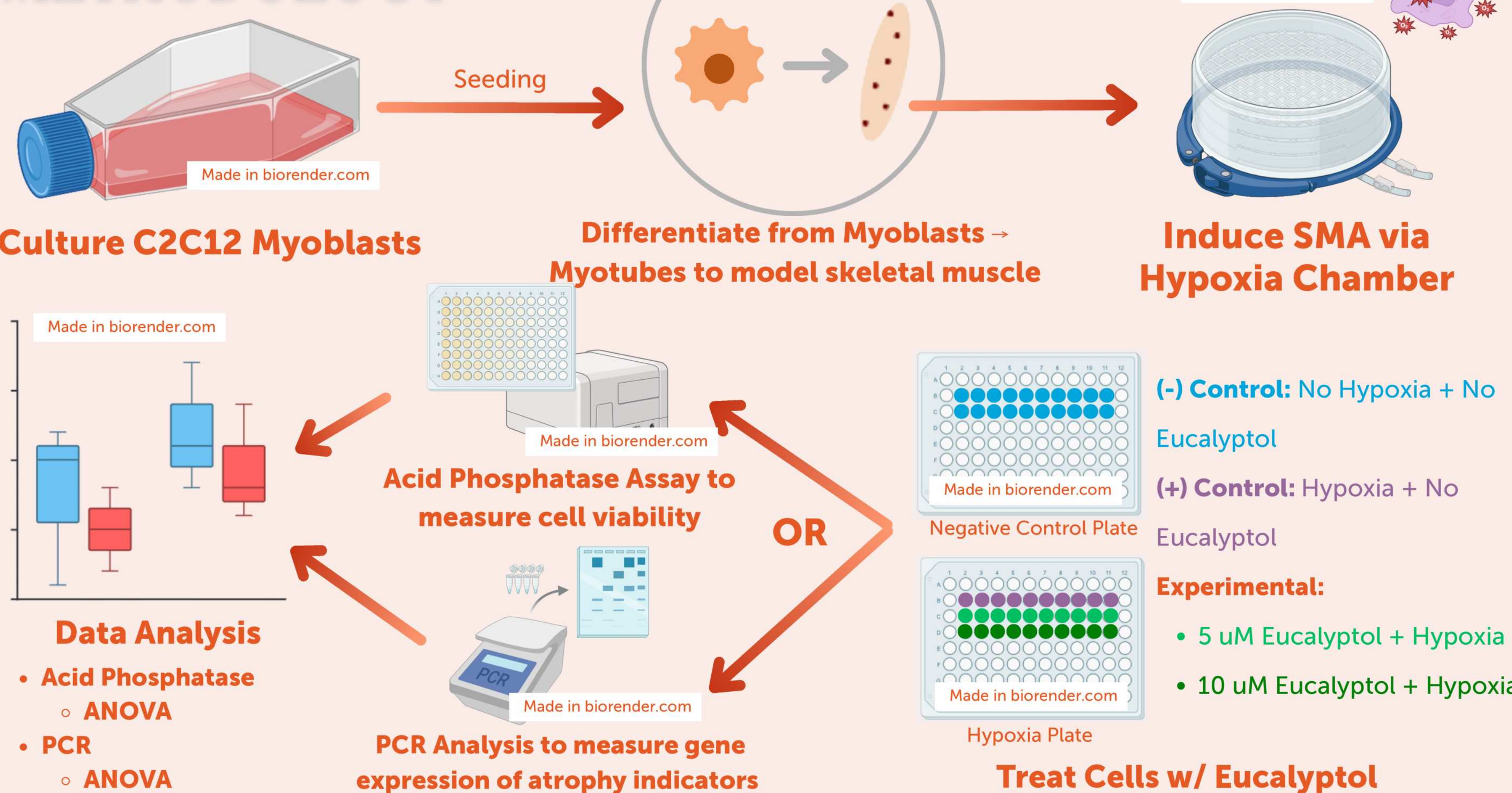
METHODOLOGY (attempted)



RESULTS



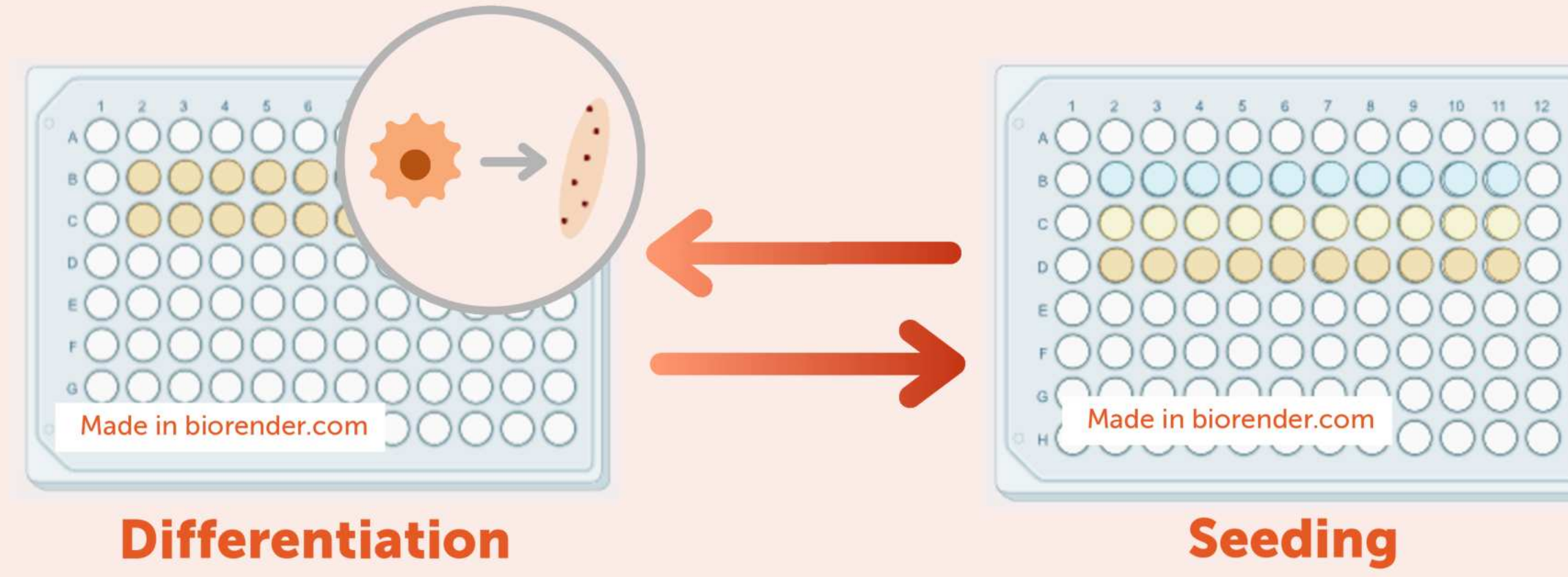
METHODOLOGY (proposed)



DISCUSSION

ACID PHOSPHATASE

- Errors in seeding → Data is an **inaccurate representation**
- Differentiation was done **before seeding** → Cells prone to Trypsin
- Procedure **shifted** to differentiate the cells **after** seeding to prevent cell damage



- **Unknown** whether or not the original hypothesis is supported due to inaccuracies in the data

PCR

- Wasn't feasible to run PCR on few cells
- Errors in seeding prevented completion

FUTURE DIRECTIONS

- Eucalyptol remains a **strong candidate** as an alleviant to SMA
- Determining if Eucalyptol works better, **synergistically** with another strong antioxidant compound.

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